



**THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED**

**Q.1. Answer the following short questions:**

**(6x5=30)**

- I. Systematic and Randomized Design
- II. Disadvantages of CR design
- III. Treatment and Experimental Unit
- IV. Mixed Effect Models.
- V. Assumptions of ANCOVA
- VI. Advantages of Latin Square Design

**Answer the following questions:**

**(3x10=30)**

**Q2.** A composite measure of screen quality was made on screen using four lacquer concentrations, four standing times, four acryloid concentrations (A,B,C,D) and four acetone concentrations ( $\alpha, \beta, \gamma, \delta$ ). A Greco Latin square was used for the analysis purpose and the following results were obtained. Find the relative efficiency of this design with:

- i) Latin Square Design      ii) RCB Design      iii) CR Design

| Source         | Columns | Rows  | Latin Letters | Greek Letters | Error | Total  |
|----------------|---------|-------|---------------|---------------|-------|--------|
| Sum of Squares | 25.25   | 85.25 | 4.25          | 26.25         | 10.75 | 151.75 |

**Q3.** Consider the experiment in which three different machines produce a monofilament fiber for a textile company. The process engineer is interested in determining if there is a difference in the breaking strength of the fiber produced by the three machines. However, the strength of a fiber is related to its diameter, with thicker fibers being generally stronger than thinner ones. A random sample of five fiber specimens is selected from each machine. The fiber strength (y) and the corresponding diameter (x) for each specimen are shown in the following Table. Perform the Analysis of Covariance.

| Machine 1 |    | Machine 2 |    | Machine 3 |    |
|-----------|----|-----------|----|-----------|----|
| Y         | X  | Y         | X  | Y         | X  |
| 36        | 20 | 40        | 22 | 35        | 21 |
| 41        | 25 | 48        | 28 | 37        | 23 |
| 39        | 24 | 39        | 22 | 42        | 26 |
| 42        | 25 | 45        | 30 | 34        | 21 |
| 49        | 32 | 44        | 28 | 32        | 15 |

**Q4.** An experiment was conducted to investigate the blink rates during reading. Data were recorded for four successive twenty-five-minute periods of reading. Please assume that the experiment was performed on four individuals (take them as Blocks) and four periods (take them as Treatments). The data obtained during the experiment is given below:

| Blocks | Treatments |     |     |     |
|--------|------------|-----|-----|-----|
|        | 1          | 2   | 3   | 4   |
| A      | 124        | 123 | 128 | 127 |
| B      | 118        | -   | 117 | 119 |
| C      | 141        | 141 | 149 | 139 |
| D      | 146        | 169 | 174 | -   |

Estimate the missing observations on 2<sup>nd</sup> row and 2<sup>nd</sup> column, 4<sup>th</sup> row and 4<sup>th</sup> column by Yates method. Analyze the data.